Chapter 18, Problem 3RQ

Problem

The inverse Fourier transform of $\dfrac{e^{-j\omega}}{2 + j\omega}$ is

(a) $e{-}2t$

(b) $e{-}2tu(t - 1)$

(c) $e{-}2(t - 1)$

(d) $e{-}2(t - 1)u(t - 1)$

Step-by-step solution

Step 1 of 1

We have

$$F^{-1}\left[\frac{e^{-j\omega}}{2+j\omega}\right] = \frac{1}{2\omega}\int_{-\infty}^{\infty} \frac{e^{-j\omega}}{2+j\omega} e^{j\omega t} d\omega$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{e^{j\omega t(t-1)}}{2+j\omega}$$

$$= e^{-2(t-1)}u(t-1)$$

$$= \boxed{d}$$